



Service Bulletin Number: 5504580

Released Date: 23-jul-2018

Idler Gear Bushing Inspection Procedure

Idler Gear Bushing Inspection Procedure

Product Affected

- 38 Liter - All engines
- 50 Liter - All engines

Issue

Excessive idler gear bushing wear.

Symptom

- High copper levels in lubricating oil samples. See table 1 below.
- Idler gear bushing wear causing gear train damage.

Root Cause

Corrosion and pitting on idler gear bushing caused by degraded oil.

Familiarisation

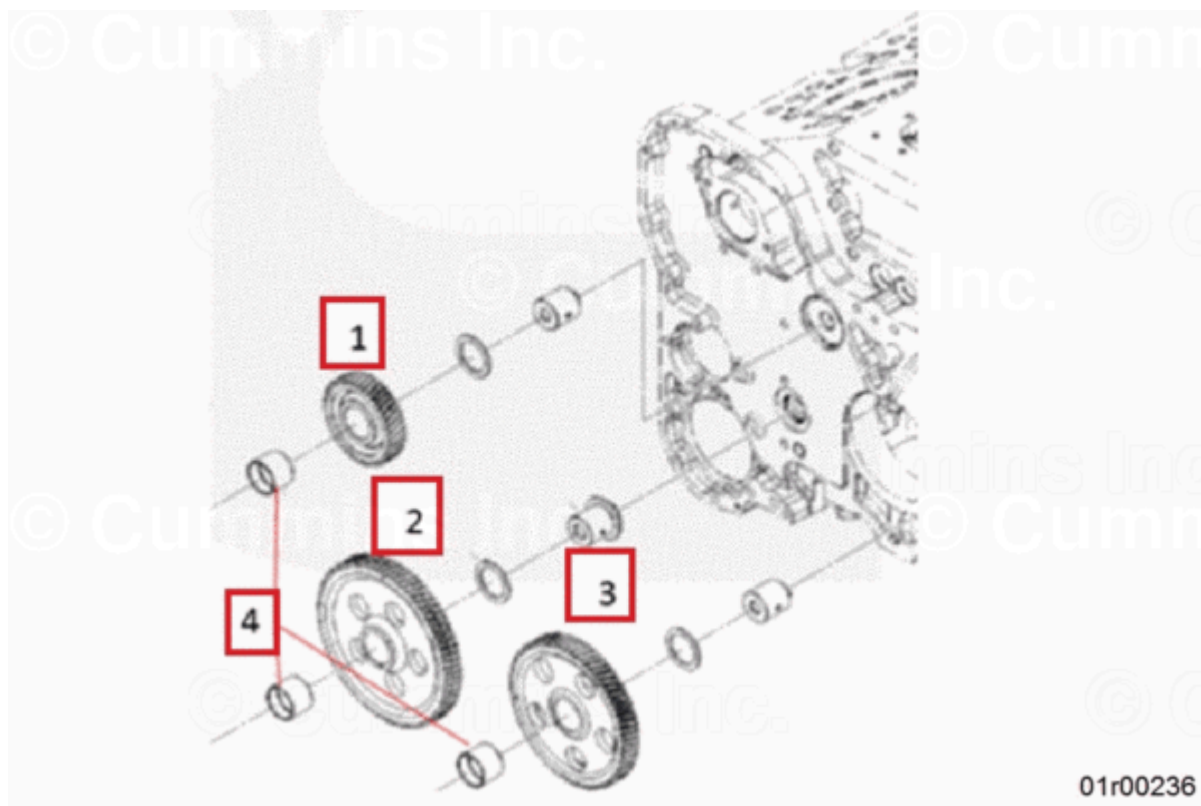


Figure 1, Component Familiarisation

Table 1, Component Familiarisation

Label Number	Part Name
1	Right Bank Camshaft Idler Gear
2	Right Bank Water pump Idler Gear
3	Left Bank Fuel & Camshaft Idler gear
4	Idler Gear Bushing

Resolution:

It is recommended that for sites experiencing excessive idler gear bushing wear, a 250 fixed-hour oil change be implemented unless extended oil analysis is performed according to Service Bulletin Number 4022060, Oil Analysis Techniques for High Horsepower Diesel Engines.

Refer to 5411406 Fluids for Cummins Products Service Manual for guidance on oil practice to avoid oil becoming corrosive.

For sites experiencing excessive idler gear bushing wear, see Table 2 below for revised copper limits and total base number limits.

Table 1, Copper Content and Acidity Limits for Lubricating Oil

	Copper (Cu) (ppm)	Total Base Number (mg KOH/g)
Caution	30ppm	9
Critical	50ppm	7

Bushing Inspection at Rebuild

Gears that meet reuse criteria **must** have bushings visually inspected in addition to meeting required tolerances. See corresponding Service Manual. Reference Service Procedure 001-063.

Figures 2, 3, and 4 represent examples of compromised idler gear bushings at different levels of severity, which are **not** suitable for reuse. If any of these examples are observed, idler gear bushing **must** be replaced or a new idler gear and idler gear bushing assembly installed.

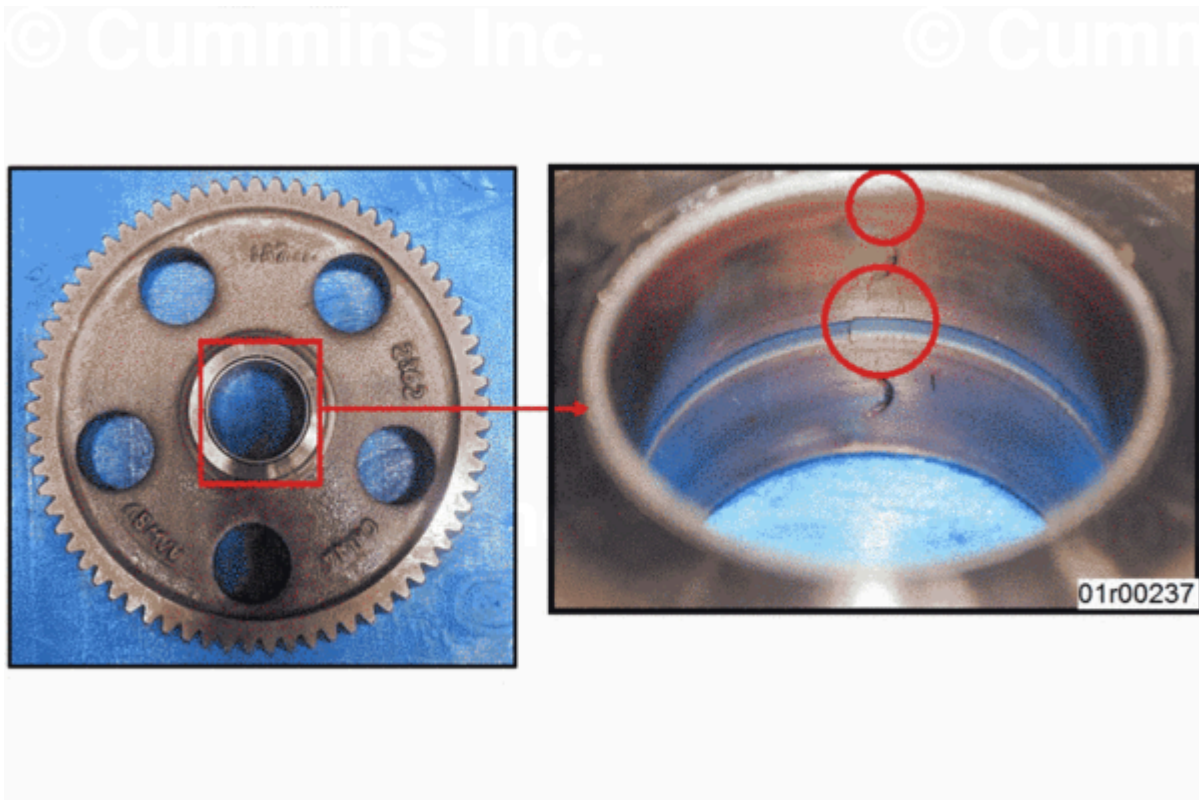


Figure 2, Material Loss Around Clinch Joint; Small Areas Of Pitting.

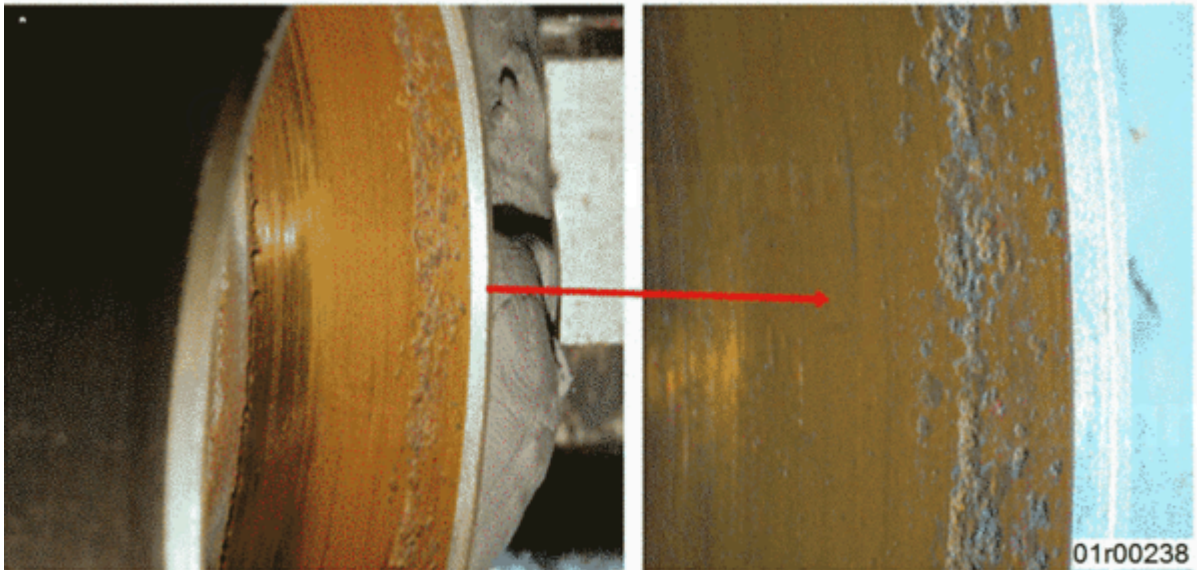


Figure 3, Heavy Wear On Oil Groove; Heavy Pitting On Outer Edge Of Idler Gear Bushing.

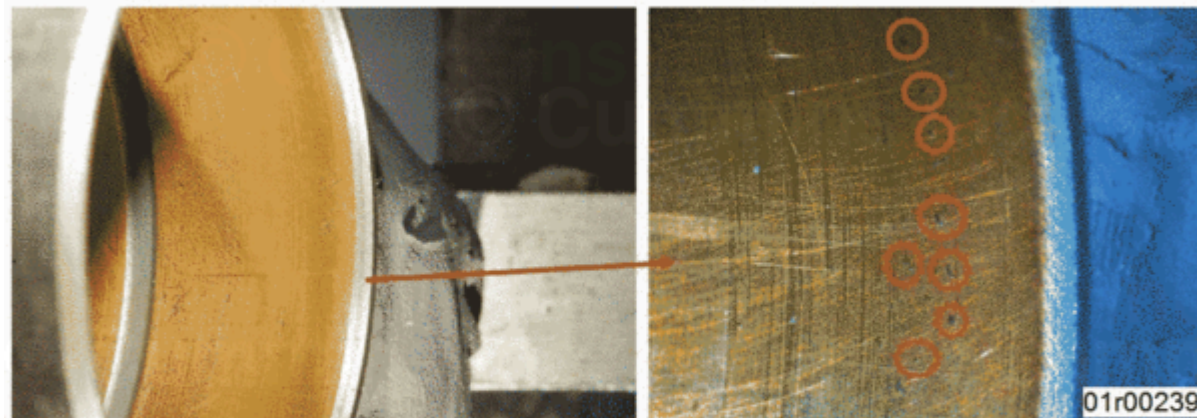


Figure 4, Micro pitting On Running Surface of Idler Gear Bushing. Idler Gear Bushings must be inspected carefully for micro pitting.

Re-bushing Criteria

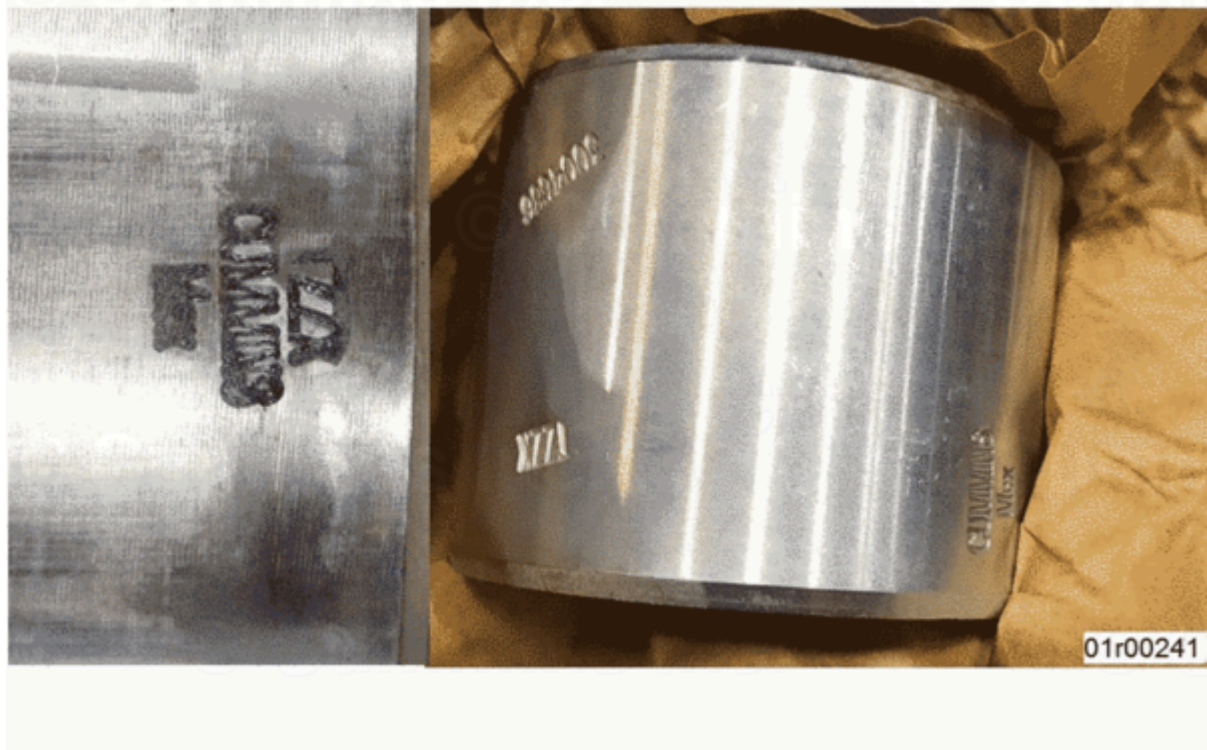


Figure 5, Genuine Cummins® Approved Stamped Idler Gear Bushings.

- Idler gear bushing **must** have a Cummins stamp. If stamp is not present, it has **not** been sourced from an approved supplier and **must not** be used.
- Idler gear bushing must also have TZZX stamp referencing supplier.
- After idler gear bushing has been replaced, it **must** be machined to meet design specification to remove tin flash.
- Once tin flash has been removed and bushing ID is within specification, idler gear **must** be installed immediately to avoid corrosion from environmental exposure.

Table

Bushing Internal Diameter	Design Spec (mm)	Service Spec (mm)
MIN	47.637	47.638
MAX	47.663	47.714

Document History

Date	Details
2018-7-23	Module Created

